A Didactic Alignment (N = 4 Taxa, L = 16 Sites) and Branch Code Anatomy

Raw Uncompressed Cost: 4 × 16 × ln 4 = 88.72 nats | Ancestral Root X_r Cost: 16 × ln 4 = 22.18 nats

Shared (1–3) | Private Substitutions (4–11; m_c=2) | Invariant (12–16)

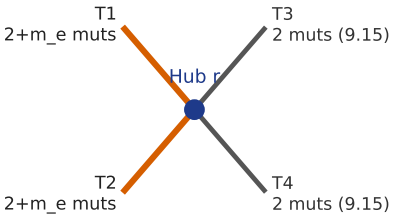
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Root X_r	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	22.18 nats
Taxon T1	G	G	G	C	C	.	.	.	.	.	.	.	.	.	.	.	m_e + 2 muts
Taxon T2	G	G	G	.	.	T	T	.	.	.	.	.	.	.	.	.	m_e + 2 muts
Taxon T3	.	.	.	.	.	.	.	C	C	.	.	.	.	.	.	.	2 priv. muts
Taxon T4	.	.	.	.	.	.	.	.	.	T	T	.	.	.	.	.	2 priv. muts

Single-Branch Code Length (Eq. 1): L_branch(m, 16) = ln 2 + L_N(m) + ln C(16, m) + m ln 3

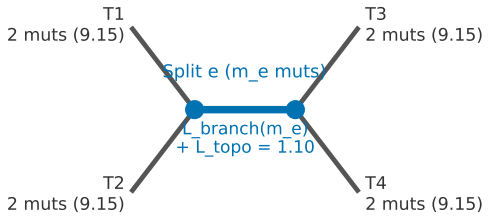
- m = 1 mut: 0.69 + 0.96 + ln(16) [2.77] + 1.10 = 5.52 nats (Unamortized new-branch startup)
- m = 2 muts: 0.69 + 1.47 + ln(120) [4.79] + 2.20 = 9.15 nats (3rd mut on leaf: +2.99 nats)
- m = 3 muts: 0.69 + 1.82 + ln(560) [6.33] + 3.30 = 12.14 nats (m = 4: 14.68; m = 5: 16.87 nats)

B Star Polytomy (T_star) vs. Resolved Quartet (T_split)

Factoring m_e shared A→G sites onto stem e saves ΔS(m_e) = 2 [L_branch(2+m_e, 16) − L_branch(2, 16)]



Unresolved Star T_star
(0 internal splits; m_e duplicated in parallel on both T1 and T2)

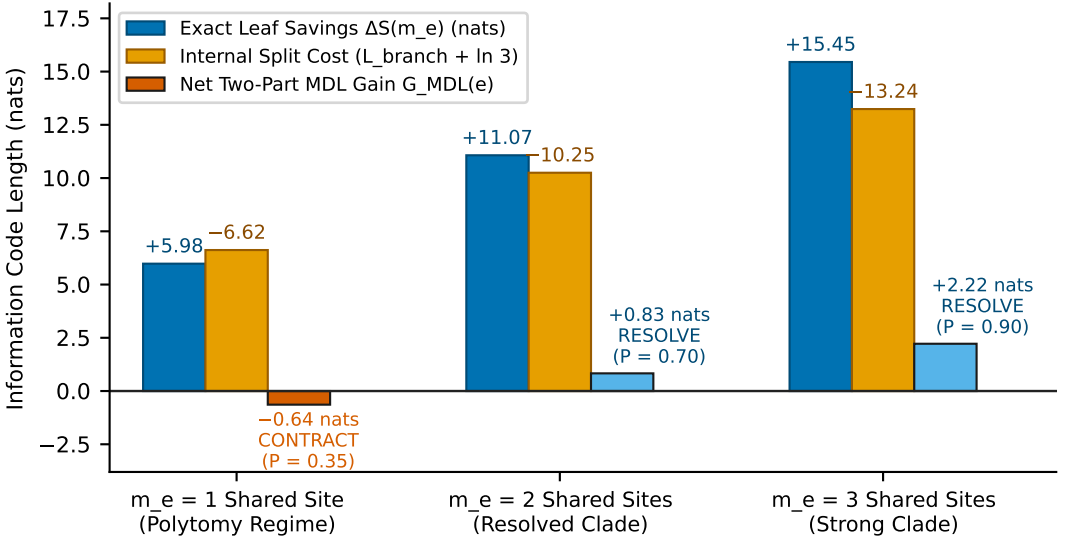


Resolved Quartet T_split
(1 of 3 quartets: L_topo = ln 3 = 1.10; shared muts encoded once on stem e)

Two-Part MDL Split Law (Eq. 5): G_MDL(e) = L_MDL(T_star) − L_MDL(T_split) = ΔS(m_e) − L_branch(m_e, 16) − ln 3

Split e is retained iff G_MDL(e) > 0 nats (Posterior Support P(e | A) = 1 / [1 + exp(−G_MDL(e))])

C MDL Split Threshold: Why m_e = 1 Contracts while m_e ≥ 2 Resolves



D Asymmetric Branch Weights (LBA Immunity) & Lateral Shunts

1. Why Combinadic Marginal Weights Resist Long-Branch Attraction (Eq. 4)

Suppose T1 & T3 are LONG branches (m_long = 6 muts on L = 16) and T2 & T4 are SHORT (m_short = 1 mut):

- Short-branch positional weight: w_short = ln[(16 − 1)/(1 + 1)] = ln(7.5) = 2.01 nats/site
- Long-branch positional weight: w_long = ln[(16 − 6)/(6 + 1)] = ln(10/7) = 0.36 nats/site

→ Grouping the two LONG branches {T1, T3} on a chance collision saves only 2 w_long = 0.71 nats of site address, whereas grouping the SHORT branches saves 2 w_short = 4.03 nats (5.6× stronger signal).

2. When Non-Sister Lineages Share Parallel States: Lateral Shunt Peeling (Eq. 8)

Once backbone tree ((T1,T2),(T3,T4)) is fixed, suppose non-sister clades share m parallel replacements:

- Endpoint Address Header: 2 ln |V| nats to specify donor and recipient nodes
- Contiguous HGT Tract: L_mask^INTERVAL = ln 2 + 2 ln L + L_branch(d_[a,b], b − a + 1)
- Scattered Convergence: L_mask^SPARSE = ln 2 + ln C(M_pool, m) (certified iff savings > header + mask)